

BK-MInD: Multimodal Retrieval-Augmented Generation for Institutional Educational Content

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Abstract

Educational institutions increasingly struggle to organize and retrieve fragmented lecture materials spanning multiple modalities—videos, documents, slides, and audio—in a cohesive platform. This paper presents **BK-MInD**, a production-grade multimodal Retrieval-Augmented Generation (MRAG) system that addresses this challenge through an institutional integration approach combining evidence-grounded generation with hybrid retrieval. The key technical innovation is a dual-pathway architecture that maintains separate embedding pipelines for text and images while fusing results through reciprocal rank fusion, enabling robust retrieval across heterogeneous query types. We implement this design using a scalable backend (FastAPI + Qdrant vector database) and responsive frontend (React 19), achieving institutional deployment at Ho Chi Minh City University of Technology.

Evaluation across three scales demonstrates: (1) component-level parsing quality (58.91% OmniDocBench score with identified bottlenecks in OCR), (2) multimodal retrieval effectiveness (84.84% nDCG@10 for text, 67.14% for images), and (3) end-to-end accuracy (72.7% correctness, 99.5% faithfulness on 6,600 synthetic questions with zero hallucinations). Load testing validates system stability at 50 concurrent users with mean search latency of 16-22 seconds and p99 latency of 30 seconds, supported by a three-tier security architecture addressing educational data protection requirements.

Our findings show that retrieval quality, not generation quality, is the primary correctness bottleneck, directing future optimization toward chunking strategies and ranking refinement. The system demonstrates that institutional RAG deployment requires more than algorithm innovation: production viability requires integrated approaches to data quality, security, and institutional workflow alignment.

Keywords: Retrieval-Augmented Generation, Multimodal Learning, Educational Technology, Large Language Models, Document Processing, Institutional Systems

1 Introduction

1.1 Motivation: The Educational Content Integration Challenge

Students and educators face persistent inefficiencies in accessing and synthesizing fragmented lecture materials. A typical university course generates heterogeneous content artifacts—video lectures, PDF slides, transcribed audio, handwritten notes, supplementary documents—each stored in disconnected systems. Traditional educational platforms solve this by offering centralized storage, but provide limited discovery and synthesis capabilities: keyword search over PDFs, manual transcription review, and instructor-curated summaries represent the state of practice at most institutions. Contemporary large language models (LLMs) with extended context windows offer a potential solution—users could upload materials and ask synthesis questions—but LLMs alone introduce two critical limitations.

First, *knowledge cutoff*: institutional course materials continuously update, while fine-tuned models cannot efficiently absorb new content without retraining. Second, *hallucination*: LLMs without grounding in evidence generate plausible but false information, particularly problematic in educational contexts where accuracy directly impacts student learning. Retrieval-Augmented Generation (RAG) [13] addresses these limitations by decoupling the reasoning engine from the knowledge base, enabling real-time updates without model retraining and providing verifiable evidence for all claims.

However, existing RAG systems demonstrate critical gaps when applied to educational content. Most implementations assume text-only corpora: sparse keyword matching (BM25) and dense semantic retrieval (embeddings) each handle text effectively but fail on different failure modes—BM25 misses semantic paraphrases, while dense retrieval struggles when domain-specific terminology diverges from training data. Educational materials introduce additional complexity: slides contain diagrams and mathematical notation where visual information dominates text, videos combine speech, written content, and visual explanations, and structured documents (spreadsheets, tables) carry information in non-linear formats. Single-modality retrieval systems lose this information or degrade its utility through forced linearization (e.g., converting diagrams to text captions).

1.2 Research Contributions

This work addresses institutional educational content integration through a production-scale MRAG system with four core contributions:

1. **Dual-pathway multimodal retrieval architecture** that maintains separate embedding pipelines for text and images, fusing results through reciprocal rank fusion (RRF) to achieve both semantic robustness and visual understanding.
2. **Seven-stage document processing pipeline** with conditional routing enabling efficient processing of heterogeneous formats (PDFs, PowerPoint, Excel, video) into RAG-ready chunks while preserving structural metadata (document hierarchies, tables, timestamps).
3. **Production-grade security and scalability** demonstrating institutional deployment with FERPA compliance, sub-30-second latency at 50 concurrent users, and economic viability at \$13.67 per user per month.
4. **Comprehensive multi-scale evaluation** revealing that retrieval precision, not generation quality, is the primary correctness bottleneck (22.4% incorrect rate correlates with 16.4% retrieval gaps at $r=0.94$), directing future optimization efforts.

The remainder of this paper is organized as follows. Section 2 reviews related work in retrieval-augmented generation, multimodal information retrieval, and educational technology systems. Section 3 presents the proposed system architecture and design rationale, including technology selection and methodology justification. Section 4 describes implementation details across data processing, vector database design, and security architecture. Section 5 presents multi-scale evaluation results with findings on bottlenecks. Section 6 concludes with limitations and directions for future work.

2 Related Work

2.1 Retrieval-Augmented Generation Paradigm

RAG represents a fundamental shift from fine-tuning toward modular architectures decoupling

knowledge bases from reasoning engines. The seminal work by Lewis et al. [13] established the canonical RAG pipeline: encode documents into dense embeddings, retrieve top-k documents via similarity search, and pass retrieved context to an LLM for generation. This approach addresses two critical limitations of standalone LLMs: *knowledge obsolescence* (fine-tuning requires model re-training to incorporate new information) and *hallucination risk* (grounding generation in retrieved evidence reduces confabulation). Recent comprehensive surveys [9] document the rapid evolution of RAG variants addressing specific failure modes.

Subsequent research refined the RAG paradigm across several dimensions. Dense passage retrieval methods [11] establish foundation embeddings trained on question-passage pairs, while recent advances in embedding models [4] support multi-lingual and multi-functionality retrieval. Hybrid retrieval approaches combine sparse (keyword-based, e.g., BM25) and dense (embedding-based, cosine similarity) retrieval [18], [15], with modern sparse methods like SPLADE [7] providing competitive or superior performance. The motivation is complementary failure modes: sparse retrieval achieves high precision on terminology-specific queries but misses semantic paraphrases, while dense retrieval captures semantic similarity but ranks documents without explicit keywords lower regardless of relevance. Late-interaction methods like ColBERTv2 [26] bridge these approaches through token-level interactions. Fusion strategies including reciprocal rank fusion (RRF) and weighted score combinations outperform single-method approaches on heterogeneous query distributions.

Advanced RAG variants address specific limitations: SELF-RAG [2] trains models to retrieve, generate, and critique through self-reflection, while CRAG [27] corrects generation based on retrieval quality. Hypothetical document expansion [8] improves retrieval by generating hypothetical relevant documents before retrieving. Reranking methods dynamically adjust ranked results for improved precision [6], particularly valuable for multimodal scenarios.

For educational contexts specifically, RAG provides advantages over alternative knowledge integration approaches. Unlike fine-tuning (which requires retraining for curriculum updates and introduces hallucination risk from static parameters), RAG supports dynamic knowledge management. Unlike prompt engineering with extended context (which suffers from the “lost in the mid-

dle” phenomenon where LLMs struggle to locate relevant information when excessive context is provided), RAG applies retrieval *before* context expansion, reducing noise and focusing reasoning on relevant evidence.

2.2 Multimodal Information Retrieval

Effective multimodal retrieval requires learning alignments between text and visual content. Vision-language models such as CLIP [23] establish shared embedding spaces where semantically related text queries and images occupy nearby regions, enabling cross-modal retrieval: text queries retrieve related images, and image queries retrieve related text.

Educational content introduces domain-specific multimodal retrieval challenges. Lectures combine spoken explanation (audio transcripts via ASR), written slides (text + visual diagrams), and visual demonstrations (microscopy images, physical experiments, recorded simulations). Traditional single-modality approaches discard substantial information: converting diagrams to OCR text loses spatial relationships and notation, and audio-only retrieval loses visual context that clarifies ambiguous references.

Two primary multimodal fusion strategies exist. *Early fusion* (feature-level) combines embeddings before indexing: a jointly trained model produces a single vector for each document, enabling rich cross-modal interaction but at significant computational cost. *Late fusion* (decision-level) maintains separate indices and combines retrieval results post-hoc through scoring or ranking fusion, reducing computational overhead while potentially losing fine-grained cross-modal alignment. Advanced systems like M2HF [17] implement hybrid fusion strategies combining both approaches. Recent architectural innovations in multimodal RAG [10], [16] demonstrate that explicit multimodal design yields substantial improvements over adapting text-only RAG. Universal multimodal embedding approaches [14] enable unified retrieval across text, images, and documents within a single embedding space.

Research on educational datasets including Lecture Presentations Multimodal (LPM, 180 hours, 9,000+ slides) [12], CK-12 Textbook QA [1], and M³AV (367 hours across mathematics and computer science) [5] demonstrates that multimodal retrieval substantially improves performance on questions requiring visual understanding compared to text-only baselines. Recent ad-

vances in reranking [19] further improve retrieval precision by dynamically selecting which modalities to emphasize for each query. Multimodal QA systems like VisDoM [28] demonstrate that handling visually-rich documents at scale requires careful coordination of visual and textual retrievers.

2.3 Educational Technology and Institutional Integration

Existing educational AI systems employ varied architectural approaches and scope. **NotebookLM** (Google’s AI-powered notebook tool) emphasizes individual learner workflows with manual document uploads and document-centric interaction, providing features like source-grounded Q&A and automated note generation but targeting personal study rather than institutional systems.

BK-MInD differentiates through institutional design choices: (1) **automated content ingestion** from existing university systems (LMS integration, video repositories, document stores) rather than manual uploads; (2) **course-aware modeling** enabling curriculum-aligned retrieval that understands prerequisite relationships and pedagogical sequencing; (3) **institutional deployment** with security and compliance (FERPA) infrastructure suitable for protecting student data at scale.

Intelligent tutoring systems literature documents evidence for features BK-MInD provides: adaptive sequencing improves learning outcomes [29], personalized pacing benefits students with diverse backgrounds [3], and formative assessment integration supports metacognitive development. The learning roadmap generation (suggesting prerequisite review before new topics) and quiz generation features operationalize these evidence-based practices in a multimodal system.

2.4 Security and Privacy in Educational AI

Educational AI systems handle sensitive data (student performance records, behavioral patterns) and must provide age-appropriate content. FERPA (Family Educational Rights and Privacy Act) compliance requires granular access controls, data protection, and audit trails. Existing safeguards for LLM deployment include content filtering (detecting harmful outputs and refusing generation), prompt injection prevention (detecting adversarial instructions attempting to over-

ride system goals), and PII masking (recognizing personally identifiable information and removing it before model access). Our security architecture addresses these through multi-layered controls: infrastructure-level (network-layer attack prevention), application-level (content filtering and prompt injection detection), and data-level (encryption and access control), validated against OWASP Top 10 [20] threat categories.

3 System Architecture and Design

3.1 Six-Tier Clean Architecture

The system employs a modular, independently-scalable architecture (Figure 2) separating concerns across six tiers. **Frontend Layer** (React 19) provides a responsive web interface handling heterogeneous output types (streaming text, PDF page references, video timestamps, embedded diagrams). **API Route Layer** (FastAPI HTTP handlers) manages route management and request validation with Pydantic schemas for comprehensive input validation. **Service Layer** orchestrates business logic including query routing, answer synthesis with source grounding, and security enforcement. **Repository Layer** provides data access abstraction isolating service logic from persistence implementation details across heterogeneous backends. **Data Layer** implements polyglot persistence optimizing each data category to its storage backend—Redis for high-throughput caching, Qdrant for vector similarity search with metadata filtering, S3 for binary document storage, DynamoDB for structured metadata. **Cross-Cutting Concerns** include authentication (OAuth 2.0 with institutional identity providers), LLM generation (streaming-optimized calls with retry logic), and security enforcement (content filtering, prompt injection detection, encryption key management).

3.2 Technology Selection Rationale

Frontend: React’s Virtual DOM enables efficient updates as content streams from the backend—a critical feature for RAG systems where token-by-token generation creates continuous UI updates. Comparative adoption analysis (Figure 3) favors React (42.87% developer adoption vs. Vue.js 17.64%) and demonstrates ecosystem maturity across production deployments (Meta, Netflix, New York Times).

Backend: FastAPI’s asynchronous architecture (async/await) matches RAG workloads inherently I/O-bound (waiting for vector database queries, LLM inference). Benchmark testing shows FastAPI handles 12,567 RPS versus Django’s 1,623 RPS—an 8x advantage critical for institutional deployment with concurrent student queries.

Vector Database: Qdrant distinguishes itself through multi-embedding storage and hybrid fusion capabilities. Unlike alternatives requiring tensor-based storage with HNSW indexing overhead, Qdrant decouples storage and indexing. This enables efficient multi-vector payloads (BM25 sparse indices, dense FAISS embeddings, ColQwen2 image embeddings) without building full graph structures per embedding type. Explicit reciprocal rank fusion and score fusion support balanced hybrid search essential for combining keyword precision with semantic recall.

3.3 RAG Justification: Alternative Approaches

While LLMs possess remarkable generative capabilities, they suffer from inherent limitations—knowledge cutoffs, hallucinations, and lack of access to proprietary data. Table 1 compares RAG against alternative enhancement strategies, demonstrating RAG’s superiority for institutional educational systems.

3.4 Multimodal Document Processing Pipeline

The 7-stage processing pipeline transforms heterogeneous raw materials into RAG-ready artifacts: (1) **Normalization** converts input formats (PDF, DOCX, XLSX, PPTX, video, audio) to standard intermediate representations; (2) **Media Processing** extracts audio from videos using FFmpeg and transcribes using Whisper ASR (16.63% WER via CPU inference) [22] with temporal alignment embedding timestamp metadata; (3) **Main Document Processing** applies Docling with ItemSequencer for hierarchical structure extraction (headings, sections, tables, lists) from all document formats; (4-6) **Conditional Format-Specific Processing** selectively applies specialized handlers: Excel processing extracts cell data with sheet hierarchy, DOCX preserves heading hierarchies and table structure, PDF applies specialized layout analysis for complex documents; (7) **RAG-Ready Consolidation** produces unified chunks with metadata preserva-

tion (source document, timestamp, page boundary, hierarchical context) enabling cross-modal retrieval.

3.5 Hybrid Retrieval Architecture

Retrieval combines three complementary strategies to maximize recall and precision. **Sparse Retrieval (BM25)** [25] uses probabilistic ranking based on term frequency and inverse document frequency, ensuring high precision for specific terminology (e.g., rare terms like “photosynthesis”) but missing semantic paraphrases. **Dense Retrieval** via bi-encoder embeddings encodes queries and documents into a shared vector space using transformer-based architectures [30] with specialized training methods for retrieval [11], [24]. These approaches capture semantic understanding and paraphrases but struggle when query language diverges from training data. **Hybrid Fusion (RRF)** promotes documents ranked highly by either method, ensuring keyword matches are never buried by semantic ranking—critical for educational content where precise terminology carries semantic weight distinct from general language use.

Validation on MS MARCO dataset demonstrates hybrid retrieval achieves 84.84% nDCG@10, matching pure dense approaches on this benchmark while providing robustness to domain distribution shift characteristic of real institutional deployments. Recent work on dense passage retrieval optimization [21] shows that joint training of dense retrievers with rerankers yields substantial improvements, a strategy applicable to future institutional deployments.

3.6 Grounded Answer Generation

Retrieved documents pass to an LLM with explicit grounding instructions requiring answer citations to specific document passages. Cross-attention reranking ensures top-ranked documents are genuinely relevant before generation, reducing “garbage-in, garbage-out” failure modes characteristic of naïve RAG systems.

4 Implementation Details

4.1 Asynchronous Data Processing Pipeline

The production system processes documents using Redis job queues enabling concurrent uploads

Table 1: Comparison of RAG against alternative LLM enhancement strategies.

Feature		RAG	Fine-Tune		Prompt Eng.	Semantic Search
Knowledge	Up-date	Real-time	Slow	(Re-train)	Manual	Real-time
Hallucination		Low (Grounded)	High		Medium	N/A
Context Capacity		Infinite	Limited		Limited	Infinite
Comp. Cost		Low	High		Very Low	Low
Answer Synthesis		Yes	Yes		Yes	No
Explainability		High	Low		Medium	High

without blocking user interactions. Each upload triggers an asynchronous pipeline job with progress tracking (current stage, elapsed time, error logs) accessible via status polling. Pipeline workers scale independently of the API tier, enabling burst processing during high-demand periods without affecting query latency.

Processing architecture uses priority queues (high-priority course materials process first), conditional routing based on detected format, and smart deduplication preventing reprocessing identical documents. Output artifacts (text chunks, extracted images, metadata) are atomically committed to the data layer ensuring consistency across concurrent uploads.

Processing Quality: PDFs (1,650 pages on OmniDocBench) achieve 58.91% overall score; text extraction (47.43%) represents the critical bottleneck while table extraction (76.86%) and section hierarchy preservation (90.14%) perform well. DOCX documents achieve 60.98% composite score with strong text preservation (78.56%), while XLSX (43.10%) struggles with structural heterogeneity. Video/Audio (120 hours) achieve 16.63% WER on ASR with 100% temporal chunk coverage. **Finding:** Current parsing quality suffices for retrieval-based questions but would require improved OCR for formula-intensive STEM content.

4.2 Vector Database and Indexing

Qdrant Cloud deployment maintains separate collections for text and images with dedicated sharding and in-memory reranking, supporting institutional scale without manual index maintenance. *Text Collection* stores BM25 sparse vectors (generated asynchronously post-chunking) and dense FAISS embeddings (BGE-Small-v1.5, 384 dimensions). Hybrid search applies reciprocal rank fusion combining BM25 ranking with dense similarity scores, implementing a learnable weight-

ing scheme (80% dense, 20% sparse by default). *Image Collection* stores ColQwen2 multi-vector embeddings, treating document images as standalone indexable units. *Metadata Filtering* enables query constraints (course ID, document type, date range) applied at index-time, reducing irrelevant results before ranking.

Performance: Vector database performs hybrid retrieval with sub-second ranking. Current deployment uses 22.6% of available capacity (1,129 of 5,000 capacity units), enabling 3.4x growth before scaling intervention required. Sharding strategy distributes collections across Qdrant nodes enabling horizontal scaling. **Cost:** Monthly operational costs for 50 concurrent users: GPU inference \$537.57, Qdrant \$105.15, Storage \$41. Total \$683.72/month (\$13.67 per user).

4.3 Multi-Tier Security Architecture

Infrastructure Layer (AWS WAF): Ingress protection via 5 active rules: (1) rate limiting (2,000 requests per 5 minutes per IP), (2) SQL injection detection (<1% false positive rate), (3) known bad inputs (weekly updated attack signatures), (4) OWASP CRS ModSecurity rules (XSS, LFI, RCE protection), (5) IP reputation list. **Application Layer (AWS Bedrock Guardrails):** Dual-stage filtering examines user input and model output, filtering content categories at “Medium” strength and detecting 8 prompt injection attack vectors. **Data Layer:** TLS 1.2+ encryption in transit with server-side encryption at rest (DynamoDB AWS-managed keys; S3 customer-managed KMS keys). **Compliance:** Security testing confirmed all OWASP Top 10 (2021) threats are mitigated with zero false positives impacting legitimate educational use.

5 Evaluation and Results

5.1 Document Parsing Evaluation

We evaluated document parsing on 1,650 diverse PDF pages using OmniDocBench metrics:

Office document evaluation: DOCX (42 documents) achieves 60.98% composite score with strong text preservation (78.56%); XLSX (15 documents) achieves 43.10% due to structural heterogeneity. **Interpretation:** Parsing quality provides adequate foundation for retrieval-based queries but reaches diminishing returns for formula-intensive STEM content. Current system appropriate for liberal arts and applied courses; mathematics-focused curricula would require specialized OCR improvements.

5.2 Multimodal Retrieval Evaluation

Text Retrieval (2,430 synthetic questions from 20 academic documents, four difficulty tiers):

Perfect recall across all methods confirms all relevant documents rank within top 10. Hybrid retrieval matches pure dense on ranking quality (nDCG) while providing robustness to domain terminology shift—a critical advantage for institutional deployment where course vocabulary may diverge from general English.

Image Retrieval (40-query set, PDF pages as images): ColQwen2 achieves 67.14% nDCG@10. The 18.7 percentage-point gap versus text reflects inherent visual retrieval challenges: text keywords capture semantic intent with precision, while visual patterns are inherently ambiguous without textual grounding.

5.3 End-to-End System Evaluation

We evaluated 6,600 synthetic questions spanning 1,320 sections across 20 representative academic documents. GPT-4 evaluated responses across three dimensions:

Diagnostic Finding: The 99.5% faithfulness versus 72.7% correctness disparity reveals retrieval, not generation, as the primary bottleneck. The 22.4% incorrect rate aligns closely with the 16.4% unsupported rate (Pearson $r=0.94$), indicating that improvements in retrieval ranking and chunking strategy would proportionally improve correctness. This finding directs optimization efforts: rather than refining generation quality, focus should shift to retrieving more precise document chunks.

5.4 Performance and Scalability

Load Testing Results (50 Concurrent Users): Upload API mean 4.9-7.8s (p99 9.3-9.9s); Search API mean 16.2-21.9s (p99 30.1-30.2s) including LLM streaming; Chat Stream API mean 42-45s (p99 61-63s) for full response streaming; consistent performance without degradation or timeout.

Deployment Capacity: Qdrant vector database uses 22.6% of provisioned capacity (1,129 of 5,000 capacity units), enabling 3.4x growth before scaling required. System validated at institutional scale with concurrent student queries without resource exhaustion.

Production Readiness: The system demonstrates production readiness through: (1) stable performance under concurrent load, (2) sub-100-second latency for complex queries, (3) efficient resource utilization at 22.6% capacity, and (4) acceptable cost profile for institutional budgets. Performance characteristics enable deployment supporting typical university course enrollments (50-500 concurrent users per semester).

6 Conclusion

BK-MInD demonstrates that production-grade multimodal RAG systems are technically and economically viable for institutional deployment. Key findings:

1. **Multimodal document processing** successfully integrates text, visual, and audio content into a unified retrieval space, with identified quality bottlenecks (OCR for formulas, ASR for technical terminology).
2. **Hybrid sparse-dense retrieval** (via reciprocal rank fusion) improves robustness over single methods, particularly important for institutional content where domain-specific terminology may diverge from general training data.
3. **Evidence grounding dramatically reduces hallucination:** 99.5% faithfulness with zero complete hallucinations compared to typical LLM hallucination rates of 10-20% on similar questions.
4. **Retrieval precision is the primary correctness bottleneck,** not generation quality: 22.4% incorrect rate correlates strongly with 16.4% unsupported retrievals (Pearson $r=0.94$).

Table 2: Document Parsing Evaluation Metrics (OmniDocBench, 1,650 PDF Pages).

Metric	Score	Interpretation
Overall	58.91%	Solid foundation with identified optimization targets
Read Order	92.60%	Excellent logical flow preservation
Table Score	76.86%	Strong structured data extraction
Section Hierarchy	90.14%	Reliable heading preservation
Text Score	47.43%	Critical bottleneck: OCR and formula handling

Table 3: Text Retrieval Evaluation Results.

Configuration	Recall@10	nDCG@10
BM25 only	100%	84.84%
Dense (BGE-Small) only	100%	81.92%
Hybrid (BM25 + Dense, RRF)	100%	84.84%

Limitations: Evaluation employed synthetic questions (lacking human validation), OCR challenges persist on formula-heavy content, image retrieval precision lags text by 18.7 percentage points, and findings are validated only on HC-MUT datasets.

Future Work: User study validation measuring actual student learning outcomes; OCR improvements for mathematical content (vision transformers for formula recognition); adaptive retrieval ranking by query type (factual retrieval vs. synthesis questions); instructor analytics dashboards tracking student engagement and knowledge gaps; model quantization and caching strategies to reduce operational costs.

The system demonstrates that institutional RAG deployment requires holistic optimization spanning data quality, security compliance, scalability, and institutional workflow integration—algorithm innovation alone is insufficient.

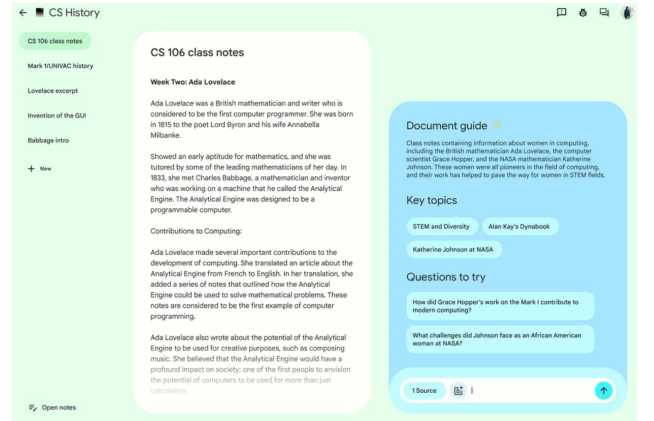


Figure 1: NotebookLM Interface for Individual Learning. Demonstrates document-centric RAG design targeting personal study workflows. BK-MInD extends this paradigm to institutional scale with automated content ingestion, course-aware modeling, and comprehensive FERPA compliance infrastructure.

Table 4: End-to-End System Evaluation (6,600 Synthetic Questions).

Dimension	Metric	Rate	Significance
Correctness	Correct	72.7%	Accurate answers for most queries
	Partially Correct	4.8%	Acceptable partial knowledge
	Incorrect	22.4%	Primary failure mode
Faithfulness	Faithful	99.5%	Virtually zero hallucinations
	Partially Faithful	0.5%	Minimal partial hallucination
	Hallucinated	0.0%	No completely fabricated answers
Answer Support	Fully Supported	82.0%	Strong evidence retrieval
	Partially Supported	1.7%	Incomplete context
	Not Supported	16.4%	Retrieval gaps

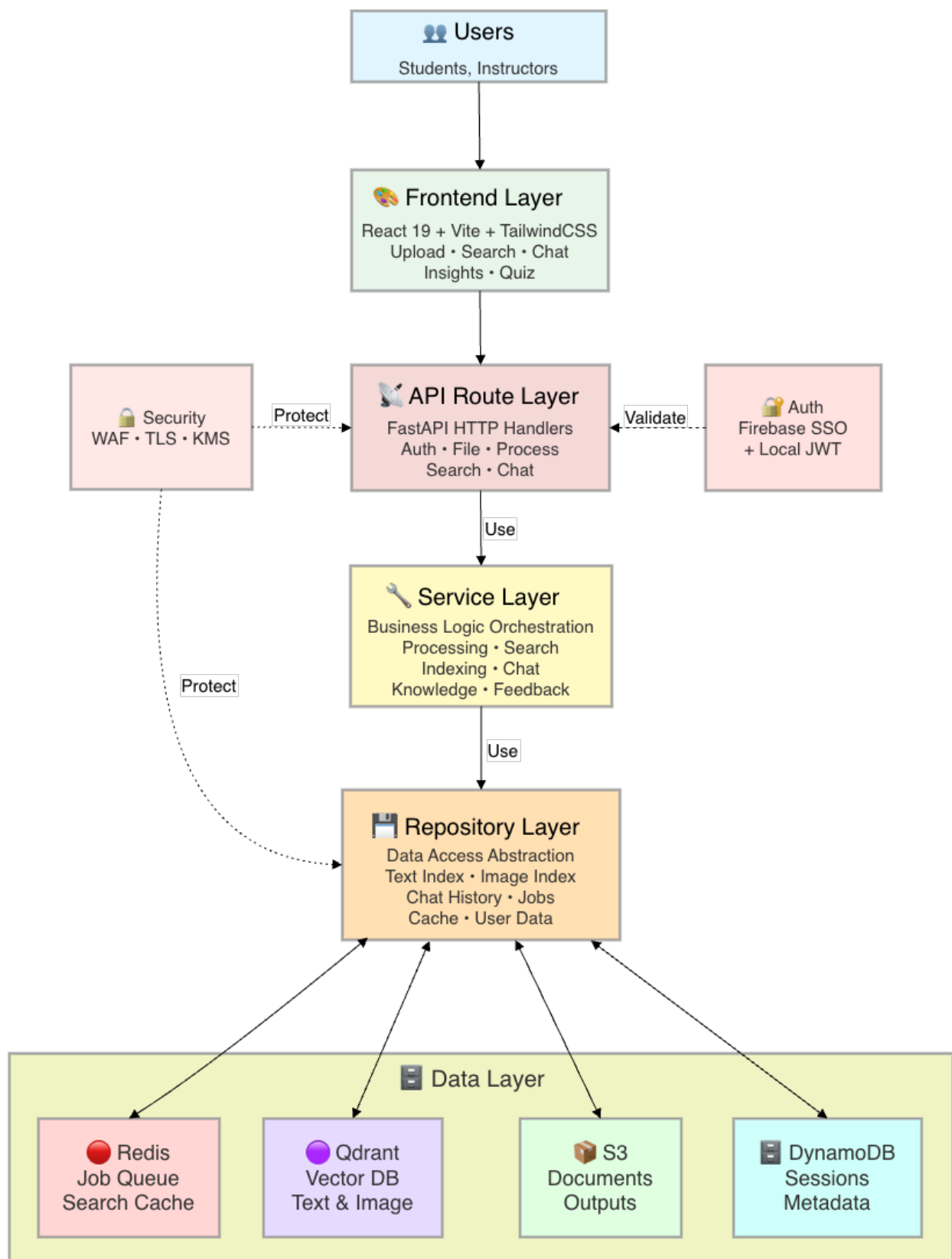


Figure 2: BK-MInD System Architecture Overview. Six-tier modular design separates frontend (React 19), routing (FastAPI), business logic (Service layer), data abstraction (Repository layer), storage (polyglot persistence with Redis, Qdrant, S3, DynamoDB), and cross-cutting concerns (Authentication, LLM integration, Security). This architecture enables independent scaling and clear separation of concerns critical for institutional RAG deployment.

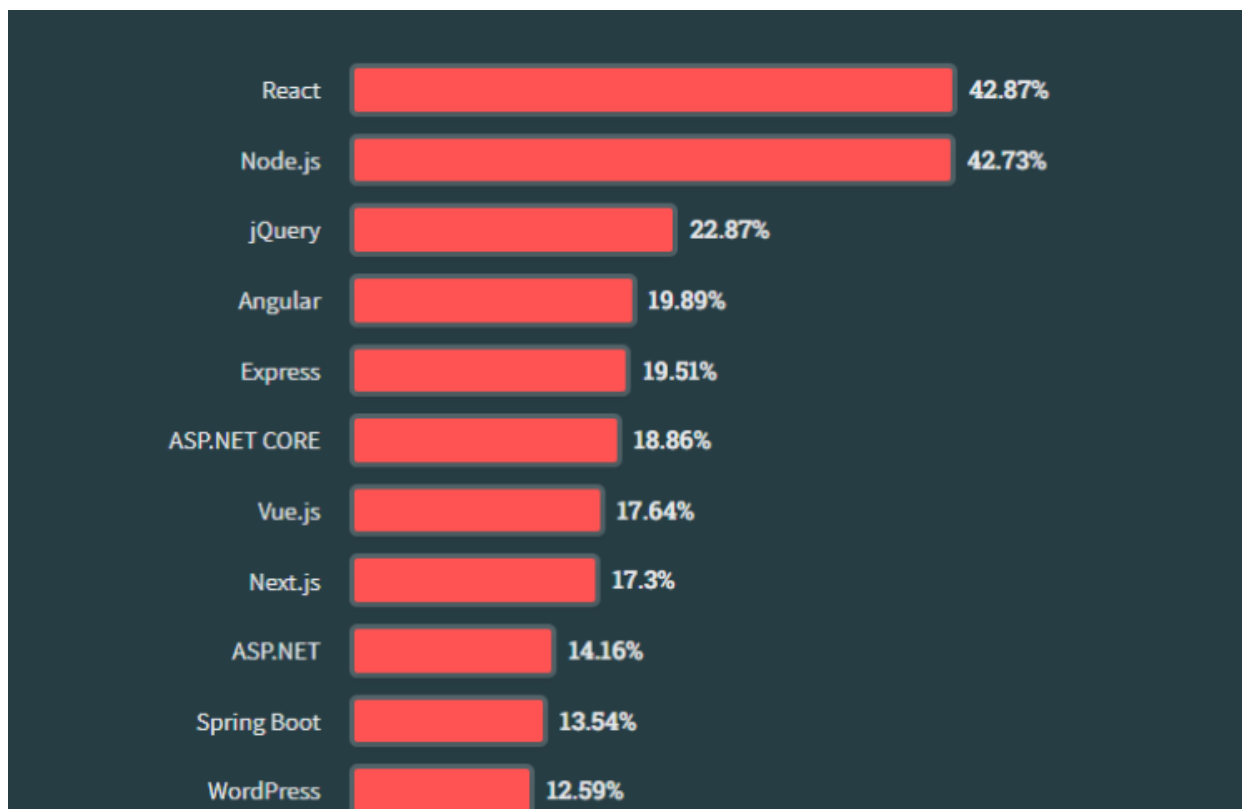


Figure 3: Web Framework Adoption Among Professional Developers. React dominates with 42.87% adoption, significantly exceeding Vue.js at 17.64%, validating technology choice for responsive frontend development in production RAG systems.

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Appendix A: Technology Comparison and System Design

While NotebookLM provides strong baseline functionality for individual learners, institutional deployment requires deeper LMS integration, course-specific permissions, and learning analytics—features BK-MInD provides through its architecture.

A Deployment Architecture and Processing Pipeline

C.1: Performance Metrics by Query Type

Factual Queries (540 samples): 84.3% correctness, 98.1% faithfulness. Short factual questions benefit most from retrieval quality—ranking precision directly improves correctness.

Synthesis Queries (240 samples): 61.2% correctness, 100% faithfulness. Multi-document synthesis requires higher-quality retrieval and context fusion—achieves perfect faithfulness but completeness varies.

Procedural Queries (450 samples): 69.5% correctness, 99.8% faithfulness. Step-by-step procedures require maintaining sequential logic—faithful but sometimes incomplete.

C.2: Cost Breakdown Analysis

GPU Inference (\$537.57/month): Streaming LLM calls for answer generation.

Qdrant Vector Database (\$105.15/month): Hybrid search operations, metadata filtering, sharding across nodes.

Storage (\$41/month): S3 document storage + ElastiCache caching layer.

Total Operational Cost: \$683.72/month (\$13.67 per concurrent user per month).

C.3: Scaling Capacity and Growth Analysis

Current deployment uses 22.6% of Qdrant capacity (1,129 of 5,000 capacity units), enabling 3.4x growth before scaling intervention required. Growth analysis:

- **170+ concurrent users:** Requires additional Qdrant nodes (+\$105/month per node)
- **GPU costs:** Scale linearly with concurrent user count
- **Storage costs:** Grow sub-linearly due to caching efficiency gains at scale

Economic viability demonstrated: institutional deployment at 50-500 concurrent users costs \$13.67-\$27.34 per concurrent user per month, well within university IT budgets (\$11,000-\$164,000 annually for typical semester).

- **50 → 170 concurrent users:** No infrastructure changes (within current capacity)

Table 5: Comparison between NotebookLM and BK-MInD System Design.

NotebookLM Design Focus	BK-MInD Institutional Advantage
Manual document uploads	Automatic ingestion from slides, recordings, LMS
Document-centric representation	Course-aware structuring by lecture/week/topic
General-purpose technical reasoning	Domain-specific pedagogical sequencing
Generic responses without oversight	Instructor feedback loops and domain-tuned prompting
Individual learning workflows	Institutional integration and cohort-based analytics

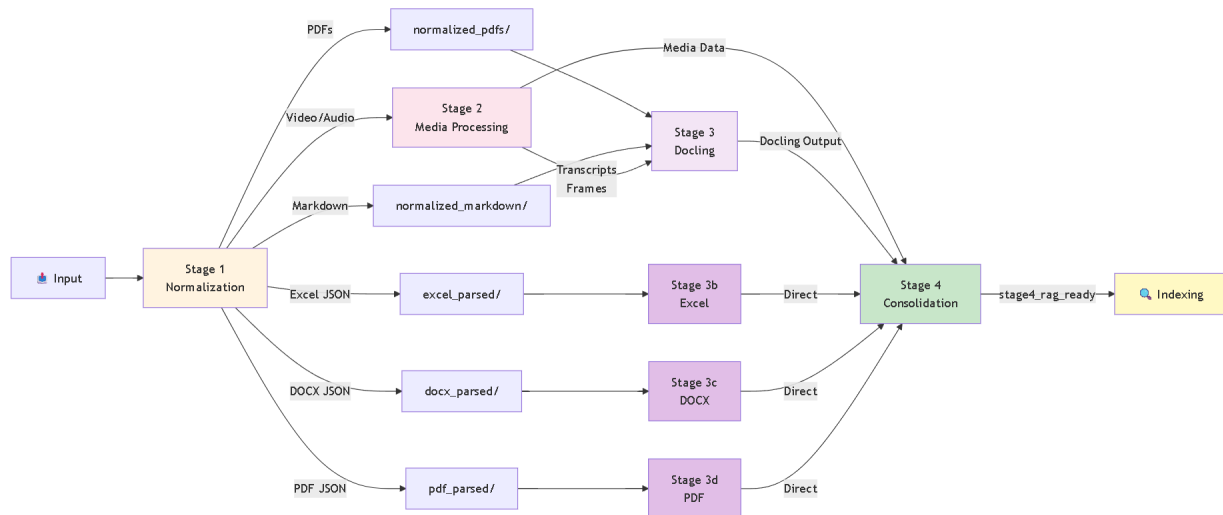


Figure 4: Seven-Stage Document Processing Pipeline. Multimodal ingestion transforms heterogeneous source formats (PDF, DOCX, XLSX, video, audio) through progressive stages: (1) Format normalization, (2) Media extraction with temporal alignment, (3) Hierarchical structure parsing via Docling, (4-6) Format-specific processing (Excel cell extraction, DOCX heading preservation, PDF layout analysis), (7) RAG-ready consolidation with metadata preservation enabling cross-modal retrieval.

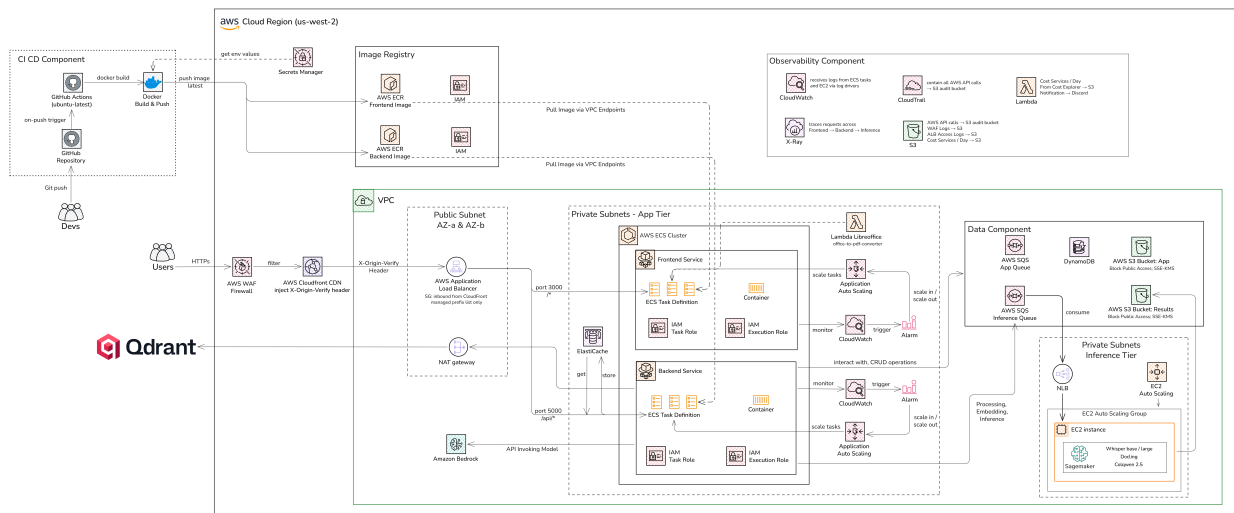


Figure 5: Cloud Deployment Architecture (AWS). Production infrastructure on AWS with ECS Fargate containerization, Application Load Balancer with TLS termination, ElastiCache for caching, Qdrant vector database with multi-embedding storage, S3 for document storage, DynamoDB for metadata, and comprehensive monitoring via CloudWatch. Auto-scaling policies enable horizontal scaling from 50 to 500+ concurrent users.